Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Problems on Lagrange's method of undetermined multipliers

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Problems on Lagrange's method of undetermined multipliers

1 Problems on Lagrange's method of undetermined multipliers

Study focus: Problems on Lagrange's method of undetermined multipliers. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000246 · EV0000247

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will small problems on Lagrange Method.

Key details

- Find the dimensions of the rectangle box open at the top of maximum capacity whose surface is 432 square centimeter.
- Two XY plus two Y Z plus two Z because the boxes open at the top.
- Because we need to find out the dimensions that is XYZ of this rectangular box.
- This volume xyz becomes function for the problem and xy plus 2yz plus 2 z equals 432 becomes constraint for the problem.
- plus 2 zx minus 432 equals 0.
- That is xy plus two yz plus two zx minus four thirty two.
- Next, as per the procedure, we need to compute the partial derivative of f with respect to x, y, as well as z.
- Then we equate fx to zero.
- That leads to yz lambda of y plus two z equals zero.
- of f with respect to y equals zero gives x plus lambda x plus two z equals zero.
- We see that 2z lambda of x minus y equals 0.
- Because 2z lambda is not zero, we can equate x minus y equals 0, then that gives x equals y.
- Similarly, multiply 2 by y and 3 by z and subtracting we get x lambda of y minus 2 z t equals 0 because x lambda not zero we can equate y minus 2 z to 0 that gives y equal to 2 z.
- Substitute x equals y, y equals to z in the constant and for you have x y z.

- We see that.
- 3x square equals 432 or x equals 12.
- Because y is x, so y is 12.
- Because z is y by 2, z becomes 6.
- So this is how we can solve problem using Lagrange method.

Example or application from the lecture

- So let's look at the first problem.
- Let us use the grandiose function which is given by f of x y z equals f of x y z plus lambda times phi.
- Let us use this in the constraint.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +5 source units in pack.json

Core explanation · 05:00–09:59

Let's look at the second problem.

Key details

- Divide the number 24 into three parts such that the continued product of the first square of the second and the cube of the third may be maximum.
- Continued product means the product we get by multiplying together in succession.
- Then divide the number 24 into three parts.
- It means we want to split 24 into three non-negative parts.
- X plus Y plus Z minus 24.
- F to zero gives two x y z power three plus lambda zero.
- Observe carefully in all these three expressions, we can use this idea.
- In each case, lambda equals minus y square z power 3, minus 2x y z power 3, minus 3xy square z square.
- We see that minus y square z power 3 and minus 2x y z power 3.
- It means from the first two terms we can get y equals to 2x.
- So because y is 2x, y becomes 8, and because z is 3x, z is 12.
- It means the three parts are 4 and 8 and 12.
- Because this is the only condition under which the inscribed rectangular solid achieves maximum volume inside the sphere.
- And f z to 0, so we see that lambda equal to minus y z by 2x, lambda equal to minus x z by two y, lambda equals minus x by 2z.
- When we solve all these in ratios, we see that x equals y equals a.

Example or application from the lecture

- Before we solve this problem, let's understand what is the meaning of continued product.
- It means if we have three parts, let us say x comma y comma z, then the continued product of the first square of the second and the cube of the third is going to be x times y square times z power three.

- Let's say xyz, that is x plus y plus z equals 24.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Problems on Lagrange's method of undetermined multipliers transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Lagrange's method of undetermined multipliers transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00066	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...
SRC00067	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00068	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...
SRC00069	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare...

C Revision and retrieval practice